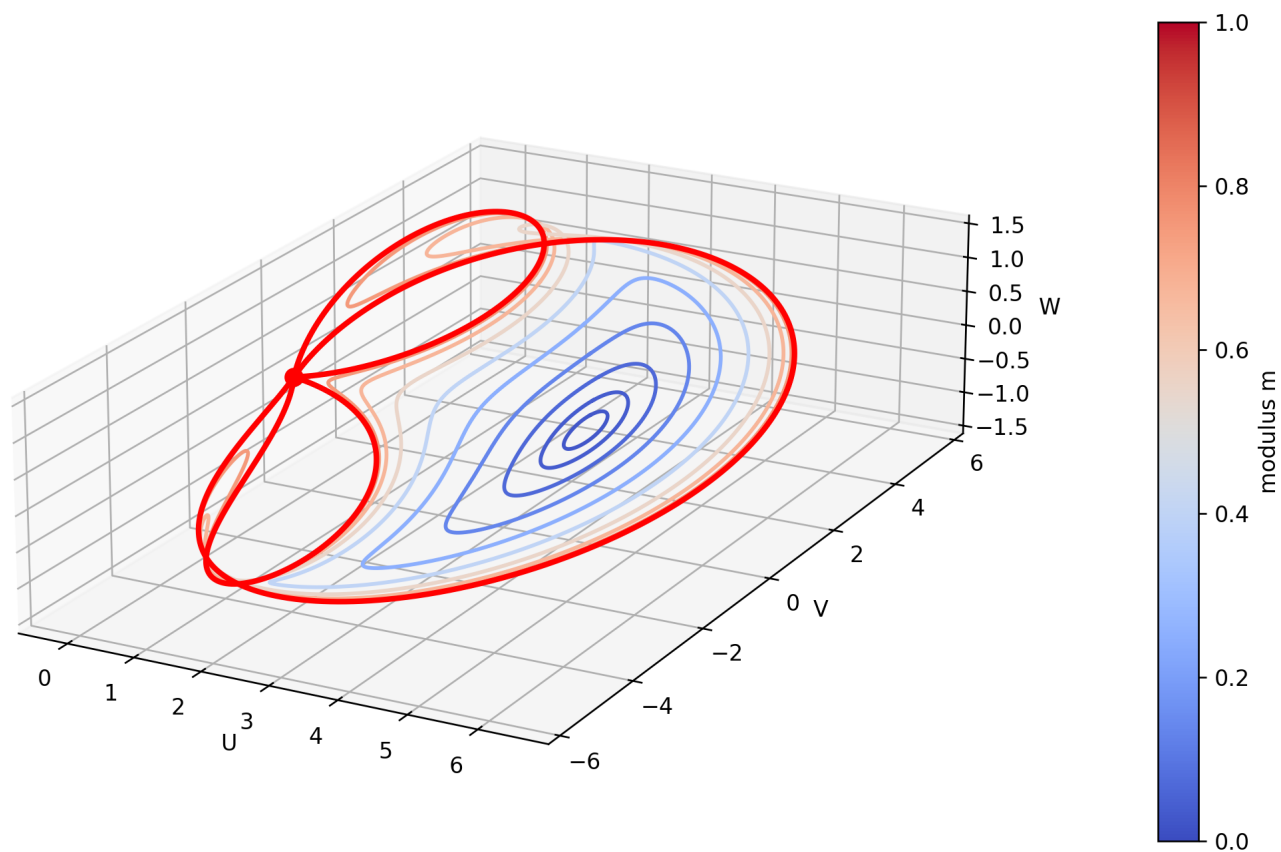


Jacobian Counter-example and Elliptic Integrals

Harm.On.ica S-O-L 5.6 (Open AI), Bradley Klee

July 2026



Abstract

A two-dimensional slice of the 2026 counterexample to the Jacobian conjecture starts with nested closed curves inside a triangle and maps the filled regions into three-dimensional space. Two areas must be kept separate. The *pre-image area* is the flat area before deformation. The *image area* is the true area of the bent and stretched sheet afterward. The pre-image area has a direct hypergeometric formula and leads, through OEIS A007004 and its derivative A006480, to Ramanujan's cubic period function. For the image area we give the full deformation density, an exact two-dimensional integral, and an exact one-dimensional integral for its derivative. Expanding the deformation density near the center produces a convergent sequence of finite linear combinations of hypergeometric functions. The first twenty area coefficients, direct numerical integration, and ordinary triangle meshes all agree. The final section poses a geometric question: can the flat triangle be folded and stretched continuously until its three corners meet, and can the given polynomial map be reached as a limit of small smooth deformations?

1 From the Jacobian conjecture to the triangle map

The Jacobian determinant records how a map changes a tiny area or volume. If it never vanishes, the map can be reversed near every point. The Jacobian conjecture asked whether a polynomial map with one fixed nonzero Jacobian determinant must also have one global polynomial inverse. In July 2026, Levent Alpöge announced a three-variable counterexample: its Jacobian determinant is constantly -2 , yet three different inputs have the same output [1, 7].

The map is

$$F(x, y, z) = (P, Q, R),$$

with

$$\begin{aligned} P &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ Q &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ R &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

The three colliding inputs lie in the plane

$$\iota(u, v) = \left(u - v, -\frac{3}{2}(u - v), \frac{27}{4}(u + v) - \frac{1}{4} \right).$$

Thus the two-variable map studied here is $G = F \circ \iota$. Its triangle corners satisfy

$$G(0, 0) = G(1, 0) = G(0, 1) = \left(-\frac{1}{4}, 0, 0 \right).$$

Filling the triangle and applying G turns the logical counterexample into a visible curved sheet. The nested curves below are level curves in that triangle and their images under G .

2 The picture before the integrals

Start with the triangle whose corners are

$$(0, 0), \quad (1, 0), \quad (0, 1)$$

in coordinates (u, v) . Inside it, consider the function

$$H(u, v) = uv(1 - u - v).$$

Its largest value is $1/27$ at the center $(1/3, 1/3)$. We label the nested closed loops by

$$H(u, v) = \frac{1 - m}{27}, \quad 0 \leq m \leq 1.$$

At $m = 0$ the loop has shrunk to the center. As m grows, it expands. At $m = 1$ it reaches the three sides of the triangle. We call the filled region inside the loop D_m and its boundary C_m .

A polynomial map G sends the starting plane into three-dimensional space:

$$G(u, v) = (U(u, v), V(u, v), W(u, v)).$$

The important geometric fact is that the three triangle corners have the same image,

$$G(0, 0) = G(1, 0) = G(0, 1) = \left(-\frac{1}{4}, 0, 0 \right).$$

Thus the final triangular boundary becomes three loops meeting at one point. For $m < 1$, the closed oval C_m becomes the space curve

$$\Gamma_m = G(C_m),$$

and the filled region becomes the mapped sheet

$$\Sigma_m = G(D_m).$$

The map used here is the restriction of the explicit polynomial map announced by Levent Alpöge in July 2026 [1].

3 Two different meanings of area

Imagine drawing a closed shape on a flat, stretchable sheet. Measure the inked region before touching the sheet: this is the *pre-image area*. Now bend and stretch the sheet according to the map and measure the same marked region along the deformed surface: this is the *image area*. The second number includes the stretching caused by the map; it is not merely a different view of the first.

Area	When it is measured	What it tells us
Pre-image area	Before applying the deformation map.	The flat area we started with; it preserves the original timing.
Image area	After deformation, measured directly along the bent and stretched sheet.	The true area after stretching; it depends on the map but not on how the finished surface is oriented.

The image area is the target of this paper. Changing the deformation map can change it, because a different map stretches the sheet differently. Rotating or reflecting the completed surface cannot change it. This combination—map dependent but orientation invariant—is exactly what we need from a quantity that belongs to the image.

4 Integral I: pre-image area

The pre-image area is the area of the filled loop before the map is applied:

$$\mathcal{A}_0(m) = \iint_{D_m} du \, dv.$$

Walking once around the boundary gives the same number:

$$\mathcal{A}_0(m) = - \oint_{C_m} v \, du,$$

with the sign fixed by the direction of travel.

In mechanics, enclosed area divided by 2π is called an *action*. Here

$$J_0(m) = \frac{\mathcal{A}_0(m)}{2\pi}.$$

The exact answer is especially simple after normalization. Define

$$F_3(m) = {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; m\right) = \sum_{n \geq 0} \frac{(3n)!}{(n!)^3} \left(\frac{m}{27}\right)^n$$

and its integral

$$S_3(m) = \int_0^m F_3(t) dt.$$

Then

$$\boxed{\mathcal{A}_0(m) = \frac{2\pi\sqrt{3}}{27} S_3(m), \quad J_0(m) = \frac{\sqrt{3}}{27} S_3(m).}$$

This is the exact quantity that preserves the original clock.

The cubic rescaling has integer coefficients:

$$\frac{S_3(27x)}{27} = x + 3x^2 + 30x^3 + 420x^4 + 6930x^5 + 126126x^6 + \dots$$

The coefficient of x^{n+1} is

$$\frac{(3n)!}{(n+1)(n!)^3},$$

which is OEIS A007004 [8].

Its derivative provides the direct link to Ramanujan. Write

$$G_7(x) = \sum_{n \geq 0} \frac{(3n)!}{(n+1)(n!)^3} x^{n+1} = x {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 2; 27x\right).$$

Then

$$G'_7(x) = \sum_{n \geq 0} \frac{(3n)!}{(n!)^3} x^n = {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; 27x\right).$$

The derivative coefficients are OEIS A006480. OEIS identifies this hypergeometric function with period functions for elliptic-curve families and connects it to Ramanujan's work on modular equations and approximations to π [9, 10]. Thus A007004 is an integrated version of the Ramanujan period sequence A006480.

The pre-image area is exact and preserves the original clock, but it does not measure how much the map stretches the sheet. It is the area before deformation, recovered from the image by undoing that deformation.

How a triangle mesh recovers the pre-image area

The mesh is a later numerical check, not the definition of the integral. A small mapped triangle with vertices $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ has an area arrow

$$\mathbf{a}_\Delta = \frac{1}{2}(\mathbf{r}_2 - \mathbf{r}_1) \times (\mathbf{r}_3 - \mathbf{r}_1).$$

Its length is the triangle's image area, while its direction records how the triangle is facing. To recover the pre-image area instead, use the correcting vector

$$B_0(u, v) = \frac{G_u \times G_v}{\|G_u \times G_v\|^2}, \quad B_0 \cdot (G_u \times G_v) = 1.$$

Then

$$\sum_{\Delta} B_0(u_{\Delta}, v_{\Delta}) \cdot \mathbf{a}_{\Delta} \longrightarrow \mathcal{A}_0(m).$$

These weights undo the local stretching of the map. They recover the area before deformation rather than measuring the size of the deformed sheet.

5 Integral II: image area after stretching

The image area is the ordinary area of the mapped sheet after the map has bent and stretched it:

$$\boxed{\mathcal{A}_{\Sigma}(m) = \iint_{D_m} \|G_u \times G_v\| \, du \, dv.}$$

This number is unchanged if the completed surface is rotated or reflected, but it changes when the deformation map changes. It therefore measures the geometry created by the map itself.

For an exact formula, use the centered coordinates

$$u = \frac{1}{3} + \frac{x}{3} + \frac{y}{3\sqrt{3}}, \quad v = \frac{1}{3} - \frac{x}{3} + \frac{y}{3\sqrt{3}}.$$

The filled region is

$$D_m = \{(x, y) : M(x, y) \leq m\}, \quad M = x^2 + y^2 + \frac{2}{3\sqrt{3}}y(y^2 - 3x^2).$$

Write

$$\mathcal{J}(x, y) = \|G_x \times G_y\| = \sqrt{C_1(x, y)^2 + C_2(x, y)^2 + C_3(x, y)^2},$$

where

$$\begin{aligned} C_1 &= -\frac{4\sqrt{3}x}{81}(24\sqrt{3}x^6y - 4x^6 - 72x^4 - 54\sqrt{3}x^2y - 27x^2 + 81), \\ C_2 &= -\frac{\sqrt{3}}{81}(48\sqrt{3}x^8y - 8x^8 - 72\sqrt{3}x^6y - 132x^6 - 108\sqrt{3}x^4y + 162x^4 \\ &\quad + 162\sqrt{3}x^2y + 297x^2 - 162), \\ C_3 &= -\frac{\sqrt{3}(2x^2 - 3)^2}{324}(48\sqrt{3}x^6y - 8x^6 - 72\sqrt{3}x^4y - 132x^4 - 108\sqrt{3}x^2y \\ &\quad + 162x^2 + 162\sqrt{3}y + 405). \end{aligned}$$

Thus the deformed area has the explicit exact form

$$\boxed{\mathcal{A}_{\Sigma}(m) = \iint_{M(x, y) \leq m} \mathcal{J}(x, y) \, dx \, dy.}$$

Nothing is hidden in the norm notation: the three polynomials above are the three components of the local area-stretching vector.

There is also an exact one-dimensional formula for the rate at which area is added. In polar coordinates,

$$M(r, \theta) = r^2 - \frac{2}{3\sqrt{3}}r^3 \sin(3\theta).$$

Let $R_m(\theta)$ be the positive solution of

$$R_m^2 - \frac{2}{3\sqrt{3}} R_m^3 \sin(3\theta) = m.$$

Differentiating the radial integral gives

$$\mathcal{A}'_{\Sigma}(m) = \frac{1}{2} \int_0^{2\pi} \frac{\mathcal{J}(R_m \cos \theta, R_m \sin \theta)}{1 - \frac{R_m}{\sqrt{3}} \sin(3\theta)} d\theta.$$

The double integral gives the accumulated image area; this single integral gives the new area contributed by the next thin shell.

At the center,

$$J_c = \mathcal{J}(0, 0) = \frac{\sqrt{6267}}{4}.$$

Normalize by

$$\Psi(m) = \frac{\mathcal{A}_{\Sigma}(m)}{\pi J_c}.$$

Then

$$\Psi(m) = \frac{1}{\pi J_c} \int_0^{2\pi} \int_0^{R_m(\theta)} \mathcal{J}(r \cos \theta, r \sin \theta) r dr d\theta,$$

and

$$\Psi'(m) = \frac{1}{2\pi J_c} \int_0^{2\pi} \frac{\mathcal{J}(R_m \cos \theta, R_m \sin \theta)}{1 - \frac{R_m}{\sqrt{3}} \sin(3\theta)} d\theta.$$

These are exact formulas for the deformed area and its derivative.

Hypergeometric form near the center

The same cubic hypergeometric functions that describe the pre-image area also organize the local expansion of the image area. Put

$$F_3(m) = {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 1; m\right), \quad S_3(m) = m {}_2F_1\left(\frac{1}{3}, \frac{2}{3}; 2; m\right).$$

Expand the exact stretching density at the center,

$$\frac{\mathcal{J}(x, y)}{J_c} = P_0(x, y) + P_1(x, y) + P_2(x, y) + \cdots,$$

where P_d is the homogeneous part of degree d . For each N , integrate the finite polynomial $P_0 + \cdots + P_{2N-2}$ over D_m . Exact reduction on the cubic curves gives polynomials $A_N(m)$ and $B_N(m)$ such that

$$\Psi_N(m) = A_N(m)F_3(m) + B_N(m)S_3(m), \quad \Psi(m) - \Psi_N(m) = O(m^{N+1}).$$

As N increases, these finite hypergeometric combinations converge to the exact image-area function near the center. Differentiation remains in the same explicit family. If

$$F_3^+(m) = {}_2F_1\left(\frac{4}{3}, \frac{5}{3}; 2; m\right), \quad F_3'(m) = \frac{2}{9}F_3^+(m),$$

then

$$\Psi'_N(m) = (A'_N(m) + B_N(m))F_3(m) + \frac{2}{9}A_N(m)F_3^+(m) + B'_N(m)S_3(m).$$

The companion file gives the exact A_8 and B_8 , whose combination agrees with the first eight nonzero coefficients of Ψ . This is a practical hypergeometric form without printing several lines of very large rational numbers in the main text.

The local series begins

$$\begin{aligned} \Psi(m) = m - \frac{8798863}{78550578}m^2 - \frac{656984015341439}{4627644978100563}m^3 \\ - \frac{5888704400396591850310}{545256281712894848663121}m^4 + \frac{9229642725727364600800502643893}{856603921733574408314213636678760}m^5 + O(m^6). \end{aligned}$$

The image-area action is therefore

$$J_\Sigma(m) = \frac{\mathcal{A}_\Sigma(m)}{2\pi} = \frac{\sqrt{6267}}{8}\Psi(m).$$

The derivative has a notable integer rescaling. With

$$C = 235651734 = 54 \cdot 2089^2,$$

the first twenty exactly computed coefficients of $\Psi'(Cx)$ are integers. The first eight are

$$\begin{aligned} \Psi'(Cx) = 1 - 52793178x - 23651424552291804x^2 \\ - 565315622438072817629760x^3 + 166133569063092562814409047590074x^4 \\ + 19584594743364404616475143307241406057420x^5 \\ + 4509774622501256294260247296713228343209292240728x^6 \\ + 1755635498235399950242562905014394392824488851953334778752x^7 + O(x^8). \end{aligned}$$

All twenty integers are included in the companion table.

Independent numerical check

The exact series, direct evaluation of the integral, and ordinary mapped triangle areas agree:

m	direct integral	20-term series	12,640-triangle mesh
0.10	6.139023	6.139023	6.137024
0.40	23.182660	23.182660	23.169230
0.70	37.237531	37.243141	37.204348

6 Folding the triangle continuously

The last question is geometric rather than computational. Begin with the flat triangle in the plane and imagine lifting and stretching it continuously until the three corners meet. The given polynomial map shows the final arrangement: three distinct corners arrive at one point while each corner still has a well-defined local sheet. What we would like is a continuous movie of how that collision can be made.

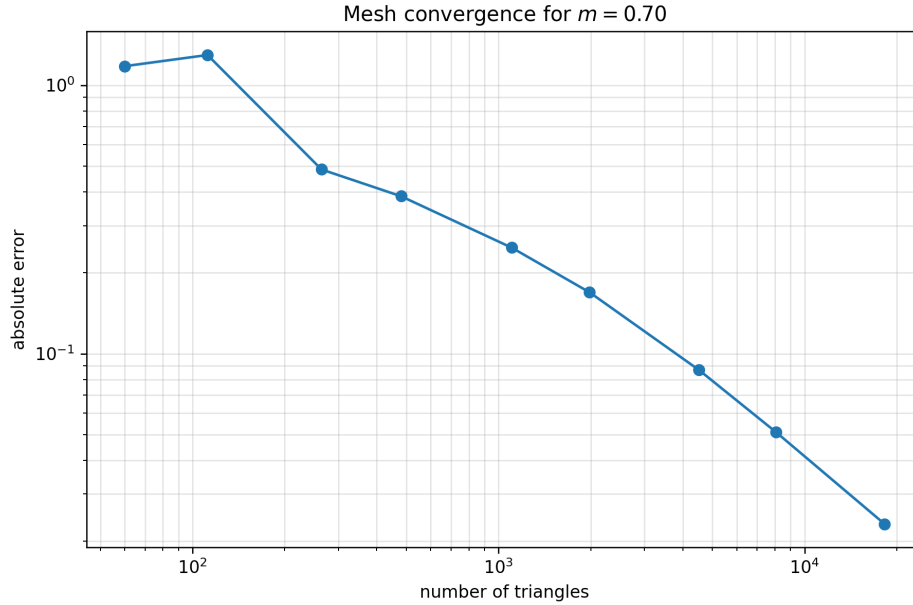


Figure 1: Ordinary triangle-area sums converge to direct numerical integration of the image area at $m = 0.70$.

One precise version is to seek maps

$$G_t : D_1 \longrightarrow \mathbb{R}^3, \quad 0 \leq t \leq 1,$$

with

$$G_0(u, v) = (u, v, 0), \quad G_1(u, v) = G(u, v),$$

after a harmless change of coordinates in the starting plane. For $t < 1$, the three corners should remain distinct and the sheet should stay smooth. At the last moment, the corners may meet exactly as they do under G .

The equilateral triangle suggests a simple picture. Lift the three corners in three different directions, as though the triangle were being folded toward a tetrahedral point, and let those directions move along curves on a sphere. The sheet is allowed to stretch, so this need not be a rigid paper-folding problem. The sphere records directions; the triangle records which points are being joined.

There are two closely related questions for colleagues:

1. Can one write down such a smooth family G_t directly?
2. Can the final map be obtained as a limit of many small, smooth changes that each stay close to the identity?

A product of ordinary constant matrices cannot do this by itself: it remains a single linear map. The useful replacement is a composition of *position-dependent* near-identity maps, for example

$$X \longmapsto X + \varepsilon V_k(X),$$

or the continuous flow

$$\frac{dX}{dt} = V_t(X).$$

Their local derivative matrices multiply along each path, while the maps as a whole can bend and stretch the triangle nonlinearly. An ambient deformation that is invertible at every stage cannot merge three

points, so the collision must occur in the final limit or in one final non-invertible step. That is not a defect; it identifies the exact event the construction must explain.

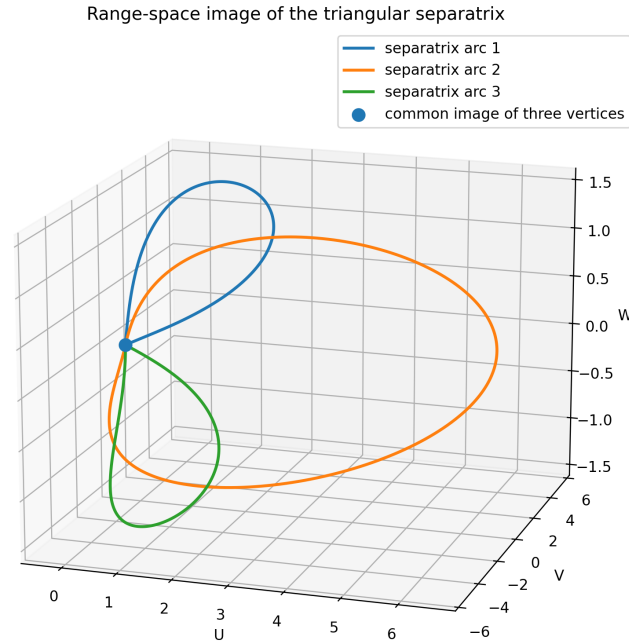


Figure 2: The three boundary arcs meet at one image point. The proposed continuous folding problem asks for a smooth path from the flat triangle to this final arrangement.

7 Files, checks, and questions for the next step

The companion files include the exact map, the three polynomial components C_1, C_2, C_3 , the first twenty image-area coefficients, the first twenty rescaled derivative integers, the $N = 8$ hypergeometric reduction, direct integration values, and mesh checks. The main files are

- `data/intrinsic_area_density.txt`;
- `data/true_surface_area_series.csv`;
- `data/image_area_hypergeometric_reduction_N8.txt`;
- `data/true_surface_area_validation.csv`;
- `scripts/derive_image_area_hypergeometric.py`.

The calculations now provide an exact deformation integral, an exact shell integral, a convergent hypergeometric reduction near the center, and three independent numerical checks. The natural next questions are whether the coefficient functions A_N, B_N approach a simpler named form, whether the integer rescaling continues to all orders, and whether a smooth family G_t can make the triangle's three corners meet in the final limit.

References

- [1] Levent Alpöge, “hello there the jacobian conjecture is false ...”, public announcement, July 19, 2026, https://x.com/__alpoge__/status/2079028340955197566.
- [2] Tom Crawford, “Oxford Calculus: Jacobians Explained,” YouTube video, <https://www.youtube.com/watch?v=YqMelRryG8U>.
- [3] Mathgeek, “The Jacobian Conjecture,” YouTube video, July 2, 2023, https://www.youtube.com/watch?v=Jxu3is_hrZI.
- [4] Lázaro O. Rodríguez Díaz, “On the origin of the Jacobian conjecture,” 2025, <https://arxiv.org/abs/2512.23614>.
- [5] “Counter Example To Jacobian Conjecture Found By Levent Alpoge,” YouTube video, July 2026, <https://www.youtube.com/watch?v=2ausgtmjn6g>.
- [6] “AI kills the Jacobian conjecture – the details,” YouTube video, July 2026, <https://www.youtube.com/watch?v=80BxOuyDBU8>.
- [7] David Speyer, “The new counterexample to the Jacobian conjecture,” Secret Blogging Seminar, July 20, 2026, <https://sbseminar.wordpress.com/2026/07/20/the-new-counterexample-to-the-jacobian-conjecture/>.
- [8] N. J. A. Sloane and James Propp, “A007004: $(3n)!/((n+1)(n!)^3)$,” The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A007004>.
- [9] N. J. A. Sloane, “A006480: $(3n)!/(n!)^3$,” The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A006480>.
- [10] Srinivasa Ramanujan, “Modular Equations and Approximations to π ,” *Quarterly Journal of Mathematics*, Vol. 45 (1914), 350–372.